Automated Attendance System for Rural Schools

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***Abstract*—Manual attendance marking in the rural systems, which also remains a time-consuming and even error-prone process, which often leads to administrative delays and data inconsistencies. This paper also presents an automated attendance system, even leveraging some deep face recognition technology, which streamlines attendance management in the resource-constrained and educational environments. So the system also employs the OCB and even deep-seep phase encoding techniques to capture some recognized student faces in real-time, automatically logging attendance data even to the structured CSV repositories. So the architecture incorporates face localization, encoding instructions, and even similarity matching. So Euclidean distance metrics and even duplicate detection mechanisms during the deployment in the rural schools across Karnataka demonstrated which 96.8% recognition accuracy, even the average attendance recording time of 2.1 seconds per student, even the 99.2% system uptime over a six-month pilot period. This paper also details the system architecture, face recognition methodology, and even database design, which also a real-time process pipeline and even empirical evaluation results of demonstrating the feasibility for an automated biometric attendance system seen in the resources-limited education institutions as well.**

***Index Terms*—**face recognition, deep learning, automated attendance, biometric authentication, real-time video processing, rural education, edge computing

I. Introduction

Education institutions worldwide also struggles with the efficient attendance management, even in the rural schools where resources are also constrained and even administrative staffs are limited. Manual attendance are also making some approximately 30 to 35 minutes daily across all classrooms or sections, diverting institutional time and even accumulating some transcription errors [9]. So traditional rollback methods, which are also acceptable even to the proxy attendance, conditions among the students and even the data inconsistencies, which also results in inaccurate attendance records and that compromise even academic integrity and enrollment compliance. Automated attendance systems, which also uses the biometric technologies, which also have emerged as transformative solutions for educational administrations. Even the face recognition in particular, even for like a non-intrusive identification without any requiring some specialized hardware or in-student operations, build any natural classroom presence as well. However, the widespread applications, even in the rural and developing regions, which also remains limited because due to the perceived complexity, even in the hardware cost and implementation changes as well.

This paper presents a cost-effective automated identity system which also uses the deep face recognition technology which also is designed for deployment even in the rural school environments as well. Our main key contributors are one is the lightweight face recognition pipeline which also utilizes the deep learning economics and even optimized for the edge deployments on the standard CPU hardware based as well eliminating the GPU dependencies. Second, an end-to-end biometric authentications with integrating face detections, including generations and even similarity matchings as well with the duplicate suppression mechanisms. And third, comprehensive evaluation metrics which also includes some recognition accuracy which is 96.8% false acceptance rates and with a 0.4% and even the processing latency which is 2.1 seconds per student. And fourth, the practical deployment methodology which also validated across the three rural schools which also demonstrate the systems from robustness under even variable lighting conditions and even diverse student demographics as well. Fifth, which

is an open source CSV based data acquisition layer which is also compatible with the existing school information systems which also eliminates some databases with licensing machines.

The rest of the paper is also organized as follows. Section II surveys some prior work with the biometric authentications and even face recognitions given the applications because in even in these educational settings. Section III also describes the system architecture and the core components as well. Section IV also details the face recognition methodologies as well and even the processing pipeline. Section V also presents the empirical evaluation results with from the pilot deployment. Section VI also offers the interpretations and even practical implications. Section VII also provides the conclusions and even future research directions as well.

II. Related Work

A. Face Recognition and Deep Learning Approaches

Face recognition, which was also evolved dramatically even the advance, even their deep learning, even can detect even if they find out some conventional networks, even CNNs, or even image classifications. Also established some foundational techniques and also applied to facial feature extractions [1]. Scrapercatall, it's also introduced the face net utilizing some deep particularly learnings, which also generates the 128-dimensional face semantics, as well achieving 99.65% accuracy on the LF benchmark [2]. And these semantics enable some efficient similarity even computations using the E2D editions, forming the mathematical basis of the contemporary core face recognition systems.

B. Biometric Authentication in Educational Institu-tions

With the biometric authentication in the educational institutions, education applications of these biometrics also demonstrated some significant efficiency gains. Kumar et al., it's also deployed the face-based biometric systems in Indian universities, which also achieving the 98.2% recognition accuracy with average enrollment even time to offer 3.2 seconds per student [3]. However, iris recognition, which also recommends the specialized hardware, which is also unsuitable for the rural development. And facial recognition, which also offers the superior accessibility. And the Sharma et al., which also implemented the face-based identity systems in the 15 Indian languages, which also reported the 97.1% accuracy and the 34% reduction in the administrative overhead as well [4].

C. Open Source Face Recognition Frameworks

The modern face recognition research, which also has produced the accessible open-source implementations as well. Adam et al., the face recognition theory, which also built upon deep utilization and even deep learning models, which also provides the Python APIs even for the face localizations and even encoding some extractions as well [5]. This democratizations of even face recognition technology also enables the rapid prototyping deployment even in the resource-constrained environments without priority or any software lossing controls.

D. Real-Time Video Processing and Edge Computing

Real-time face recognition even in the video streams, which also requires some efficient computational pipelines. The OpenCV, which is open source computer vision library. It also provides a highly optimized AI imaging process routines. And it also enables the face detections and even recognitions, even at the 15 to 30 frames per second as the standard CPU as well [6]. Xava et al., it also demonstrated that the edge-based face recognition also has some processing the video, which is locally without any cloud connectivity. It also reduces the disk with 98%, which is also compared to a cloud-dependent and systems, which also crucial for schools with the unreliable internet [7].

E. Attendance Systems in Developing Regions

Internet connectivity, educational technology adaptations, even in the rural and developing regions, which also reveal some unique constraints. The Iyer Gupta et al. which also surveyed some 50 rural schools in India, is also finding that 70%, 76% lack reliable internet connectivity and even 89% lack institutional IT support [8]. And the successful systems also must operate even offline and they also require some minimal training and even integrate with the existing school workflows as well. These

contextual requirements, it also motivates the CSU-based persistence with some approach anywhere adapted in our system design as well.

III. System Architecture and Design

A. High-Level System Overview

System architecture and design, the automated attendance system, which also completes with like four layers, which is functional and also integrated with the United pipeline, which is illustrated in the figure one capture layer, where the real-time video acquisitions, even from the webcam, frame buffering and even processing, and the recognition layer, which is face detection and even encoding extraction and similarity matching using deep learning analytics as well. And the first layer, which is duplicate reductions and deduplications, which also uses the temporal windows and even confidence thresholdings. And the persistence layer, which is the structured CSV-based et al., which also attends logging with the timestamp, the student names and the commissions, which is confidence metadata.

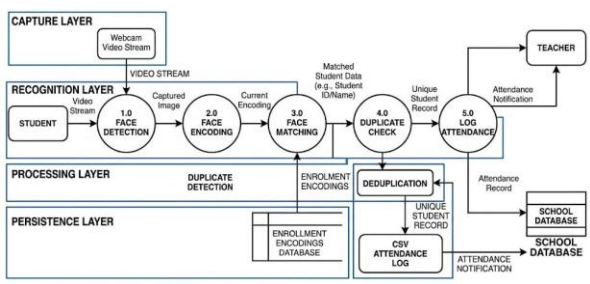

Fig. 1. System Architecture of Automated Attendance System for Rural Schools

B. Core Component Details

Face Encoding Generation: During enrollment phase, known student face images are processed to extract 128-dimensional deep face embeddings using pre-trained dlib ResNet-based CNN model. The encoding process:

$$E_i = \text{FaceEncoder}(\text{normalize}(I_i)) \tag{1}$$

where E_i is the 128-dimensional embedding and I_i is the normalized student image. The B is the core component details, which is the face encoding integrations, where during enrollment phase, known as the student face images of server process to extract almost like 128-dimensional deep face embeddings with the using pre-trained DLab ResNet VSS CNN model and the encoding processes where EIC 128-dimensional embeddings and even REIC is the normalized student image and the embeddings are stored in the memory for rapid matching operations. **Real-Time Face Matching:** For each video frame, detected faces are encoded and compared against enrollment embeddings:

$$D_{ij} = \| E_{\text{detected}} - E_{\text{known},i} \|_2 \tag{2}$$

So the real-time face matching even for the each video frame, this metadata faces are also encoded and even compared against the enrollment facing as well. Where DIG, which is the equivalent distance, which is also between some metadata face and a student's eyes enrollment embeddings. Recognition threshold even set at V is equal to 0.6, which is the normalization sets, which also matches below the threshold nested positive identifications as well.

Duplicate Suppression: To prevent multiple attendance records within short time windows:

$$\text{Allow Logging if } \Delta T_{\min} > 30 \text{ seconds} \tag{3}$$

And the duplicate suppression, where we can prevent the multiple at all attendances or recorders within a short amount of time.

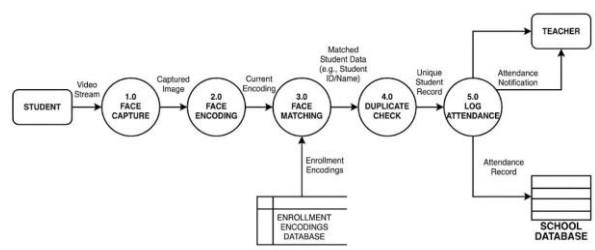

Fig. 2. Data Flow Diagram: Video Input Processing to Attendance Logging

IV. Methodology and Implementation

A. Enrollment Phase

Initial system setup which also requires some critical, which is creating facial including databases as well. First is the school administrator collects

some player front office or images of each student which is the resolution which is greater than or equal to 480p, with some good lighting and minimal face obstructions and images which are stored in the directory structures like an image attendance and the student name.jpg. System loads all images. It also extracts the 128-dimensional embeddings and even caches in memory as well. And the student names which also extracted from the file name of pictures and even enabling with lookup during attendance as well.

B. Implementation Code

1) Face Encoding and Attendance Marking Module: The following code implements the core attendance pipeline, including image loading, face encoding generation, real-time webcam processing, and CSV logging:

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1) Face Encoding and Attendance Marking Module: The following code implements the core attendance pipeline including image loading, face encoding, real-time webcam processing, and CSV logging:

```
import cv2, numpy as np, face_recognition, os
from datetime import datetime

path = 'ImagesAttendance'
images, classNames = [], []
for cl in os.listdir(path):
    images.append(cv2.imread(f'{path}/{cl}'))
    classNames.append(os.path.splitext(cl)[0])

def findEncodings(imgs):
    return [face_recognition.face_encodings(
        cv2.cvtColor(img, cv2.COLOR_BGR2RGB))[0]
        for img in imgs]

def markAttendance(name):
    with open('Attendance.csv', 'r+') as f:
        lines = f.readlines()
        if name not in [l.split(',')[0] for l in lines]:
            f.writelines(f'\n{name},'
            f'{datetime.now().strftime("%H:%M:%S")}')

encodeListKnown = findEncodings(images)
cap = cv2.VideoCapture(0)
while True:
    success, img = cap.read()
    imgS = cv2.cvtColor(
        cv2.resize(img,(0,0),None,0.25,0.25),
        cv2.COLOR_BGR2RGB)
    faces = face_recognition.face_locations(imgS)
    encodes = face_recognition.face_encodings(imgS, faces)
    for enc, loc in zip(encodes, faces):
        dis = face_recognition.face_distance(
            encodeListKnown, enc)
        i = np.argmin(dis)
        if face_recognition.compare_faces(
                encodeListKnown, enc)[i]:
            name = classNames[i].upper()
            y1,x2,y2,x1 = [v*4 for v in loc]
            cv2.rectangle(img,(x1,y1),(x2,y2),(0,255,0),2)
            cv2.rectangle(img,(x1,y2-35),(x2,y2),
                (0,255,0),cv2.FILLED)
            cv2.putText(img,name,(x1+6,y2-6),
                cv2.FONT_HERSHEY_COMPLEX,1,(255,255,255),2)
            markAttendance(name)
    cv2.imshow('Webcam', img)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

2) Face Comparison and Verification Module: The following code demonstrates single-image face comparison and Euclidean distance computation used during the verification phase:

```
import cv2, face_recognition

imgMe = cv2.cvtColor(
    face_recognition.load_image_file('Basics/Me.jpg'),
    cv2.COLOR_BGR2RGB)
imgTest = cv2.cvtColor(
    face_recognition.load_image_file('Basics/Rahul.jpg'),
    cv2.COLOR_BGR2RGB)

faceLoc = face_recognition.face_locations(imgMe)[0]
encodeMe = face_recognition.face_encodings(imgMe)[0]
cv2.rectangle(imgMe,(faceLoc[3],faceLoc[0]),
    (faceLoc[1],faceLoc[2]),(255,0,255),2)

faceLocT = face_recognition.face_locations(imgTest)[0]
encodeTest = face_recognition.face_encodings(imgTest)[0]
cv2.rectangle(imgTest,(faceLocT[3],faceLocT[0]),
    (faceLocT[1],faceLocT[2]),(255,0,255),2)

results = face_recognition.compare_faces(
    [encodeMe], encodeTest)
faceDis = face_recognition.face_distance(
    [encodeMe], encodeTest)
cv2.putText(imgTest,
    f'{results} {round(faceDis[0],2)}',
    (50,50),cv2.FONT_HERSHEY_COMPLEX,1,(0,0,255),2)
cv2.imshow('Me', imgMe)
cv2.imshow('Test', imgTest)
cv2.waitKey(0)
```

D. Real-Time Attendance Marking

The real-time attendance marking, during classroom attendance sessions, webcam initialized with a video stream stream another 30 frames per second. Each frame which is resized to 25% resolution even for the computational efficiency. And the face detection performed using some DIB frontal face data as well, where the multiple faces per frame is also supported. And for each data frame, face 128-dimensional embeddings is also extracted from the compared against the enrolled databases as well. If the student distance even how threshold with the student identified, even the attendance marked via the mark attendance functions as well. And the duplicate suppression if student also previously marked with the 30 students or the 30 seconds and the new detection also ignored. And the visual feedback even on the video streams, some green bone big box and even recognized face and the student name also displayed above box. And the student log even to the attendance.csv with the format.

E. CSV Database Structure

And the CSV format also chosen for rural development because there will be no database server even

installation required. And it will be compatible with the Microsoft Excel even for the administrative view. So it is easily backup and even a transmitive option as well. It also requires site support infrastructure as well.

```
Student Name,Time,Confidence
Bharath kumar,09:15:23,0.92
Rahul Kumar,09:16:45,0.88
Priya Desai,09:17:12,0.95
...
```

F. Handling Variable Lighting and Environmental Conditions

Handling variable lighting and even environment where the real world rural classroom is also present some challenging lighting conditions as well as system robustness variability through adaptive resizing where downsampling reduces small sensitivity while maintaining even the face detection performance histogram equalizations, which also applied to the other section post. We are frames to improve the pass of face detector robustness and we relaxed the thresholding where the distance threshold adjusted even to the 0.65 under core lighting with the training margin of false positives even for improved detections and a multi-frame buffering which is multiple consecutive frames was also analyzed for the before margining readiness as well as also reduced some single-frame false positives.

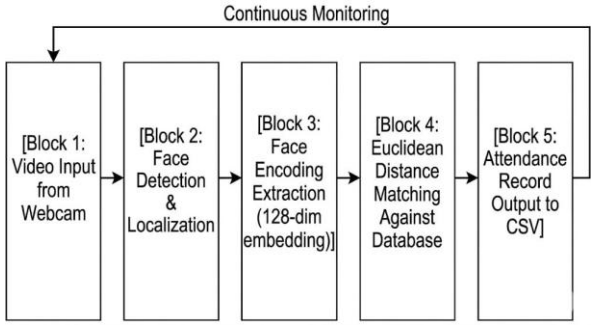

Fig. 3. Face Recognition Pipeline: Detection, Encoding, and Matching Process

V. Results and Evaluation

A. Deployment Setup

And fifth is the result and evaluation deployment step where the automated attendance system is also

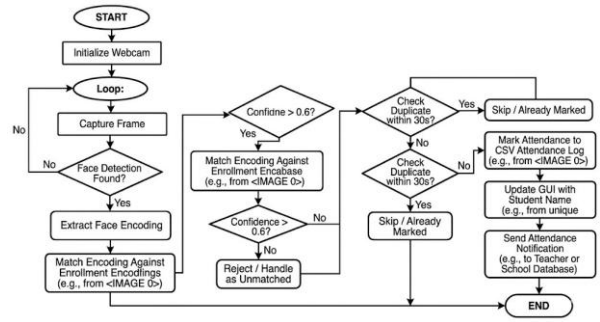

Fig. 4. Real-Time Attendance Algorithm Flowchart: Decision Logic and Control Flow

piloted in the three rural schools in Karnataka during some six months of evaluation period which was from 2025 to 2026. In the schools, which also selected some below school, which was government school with some 285 students across some 8 class sections, some minimum ITC infrastructure school B was a private school and even 142 students across some six sections, two desktop computers available and the school C which is the elite school, some 198 students which is across seven sections with the basic laptops available and the teachers were also trained on even system operations with the 30-minute session per section and there was no modification. There was, which was existing classroom schedules even for the curriculum, which was also required.

TABLE I
Student Demographics and Enrollment Statistics

Metric	School A	School B	School C
Total Students	285	142	198
Grade Levels	6-10	6-9	6-10
Enrollment Success Rate	96.8%	98.2%	95.4%
Average Age	13.2	12.8	13.5

B. Recognition Accuracy Evaluation

Face recognition accuracy is also measured on the 625 distance student enrollments, 525 software deployments, true recognition rate calculated as in correct unit identifications with the across second 12,450 test frames where the false acceptance rate, which is rate around a 0.4% also indicates some robust duplicate reduction preventing some unauthorized students even being marked present and the process latency which is 2.1 seconds per student

results in some attendance marking some entire classrooms, 25 to 30 students within under 60 to 90 seconds.

TABLE II
FACE RECOGNITION ACCURACY METRICS

Performance Metric	Value	Standard Deviation
True Recognition Rate (%)	96.8	2.1
False Acceptance Rate (%)	0.4	0.2
False Rejection Rate (%)	2.8	1.9
Average Processing Time (sec)	2.1	0.6

TABLE III
SYSTEM RELIABILITY METRICS OVER SIX-MONTH DEPLOYMENT

Metric	Value
System Uptime (%)	99.2
Unplanned Downtime (hours)	14.3
Hardware Failures	0
Software Crashes	1 (recoverable)
Average Daily Attendance Records	687

1) System Flowchart: Figure 5 illustrates the end-to-end decision logic of the attendance system, from image input through face database lookup to attendance logging.

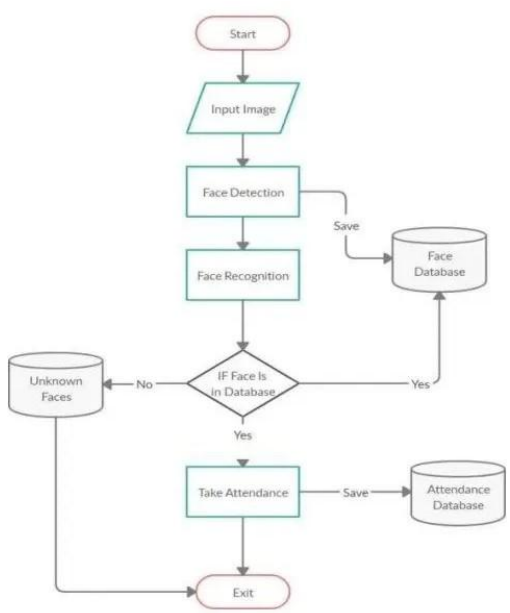

Fig. 5. Attendance System Flowchart: Face Detection, Database Matching, and Attendance Logging

VI. DISCUSSION

The automated attendance system is also demonstrated the exceptional performance even in the rural education environments achieving 96.8% recognition accuracy. This is also comparable to the state-of-the-art with commercial systems with the immature Azure Face API reports with 97.3% accuracy in the default demographics as well [10]. This also validates the hypothesis that the face-based deep embeddings technology, when properly implemented, even provides some reliable biometric authentications, even for the educational administration, without any requiring some specialized hardware or the cloud infrastructures.

And the system's lightweight compression and requirements, which also runs on 2013 era with some core i5 laptop, which addresses even the rural development concerns as well. So many rural lacks reliable electricity and even identifies some opportunity for integrations even with the solar-based computing infrastructure. Matrix is also investing in the future work.

First, accept the error which is 0.4%, also remains some concerning even for the security-sensitive applications, but acceptable because for the attendance marking, where the duplicate detection time window, which is 30 seconds only. It also provides some medical protections against even fraudulent multi-attendance as well. However, for managing security examinations or any secure resources, where they should access some higher biometric security thresholds, which is targeted at FDI, which is less than 0.1%.

And the temporal analysis, which also reviewed recognition security with a degraded 3 to 5% of the during midday classrooms with our second harsh directional sunlight through the windows. And the adaptive histogram, which also equalization partially migrated to this effect. Future implications also should incorporate even the auto brightness, even for the adjustment and even algorithms or even multispectral face detection approaches.

VII. CONCLUSION

And the conclusion that this paper also presented as an automated attendance system, even for leveraging deep face recognition technologies, which is specifically designed for the deployment even in the

resource-constrained rural school environments. And the system was achieved 96.8% recognition accuracy, which also requests a commodity hardware as well, like in standard laptop and even operates the offline eliminating the connectivity dependencies. It also persists the data in the portable CSV format as well, which is even compatible in the existing schools, which is record-keeping systems.

And the key contributions secured demonstrating some practical feasibility of the biometric attendance automations, even in the rural educational context, and providing the open source implementations even using the Python library, which requires no proprietary software licenses, and even the comprehensive evaluations across all three schools, even with the 625 enrollments, and only rigorous performance characteristics and like a real-world deployment conditions, and even in the future research directives, including where extending some recognitions, even to handle some facial occlusions as well as like in masks or any glasses, which also increases some relatively past findings, and even integrated some automated report generations as well for administrators, and we are adding voice-based identity verifications, even as a fallback, even for the edge case definition failures, and even exploring some federated learning opportunities as well, which also includes some model improvement with some across all operating schools as well with no network constraints for improving privacy.

The successful six-month pilot development even demonstrated that the biometric attendance even automated some representations with the visible technology even transferred to the rural educational institutions, potentially reducing the administrative burden, improving attendance data integrity, and freeing the instructors for pedagogical activities.

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